Spectrogram-Based CNN With Attention for Deepfake Audio Detection in Podcast Production

Brian Stever | 24042080

College of Information Science · University of Arizona

APCV 485 — Deep Learning in Cyber · Instructor: Dr. Shengjie Xu

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Problem and Motivation

- Voice cloning is now cheap and fast.
- Public episodes give attackers training data.
- A host's voice is a trust asset.
- Fake episodes threaten brand and listeners.
- Detection should sit in the workflow.
- Cybersecurity beyond networks and malware.

Research Questions

1. Can a CNN detect spoofed speech directly from a spectrogram?
2. Does the detector hold up under noise, room reverb, and MP3 compression?
3. How does it compare against other approaches? (ResNet-18, LFCC-GMM)

Primary metric. Equal Error Rate (EER).

Secondary metrics. AUC-ROC, F1, accuracy, latency.

Background

- ASVspoof: standard anti-spoofing benchmark.
- 2019 LA: clean TTS / VC attacks.
- 2021 LA: codec and channel effects.
- Squeeze-Excitation: lightweight channel attention.
- SpecAugment: time / frequency masking.
- EER: where false-accept = false-reject.

Methodology — Pipeline



- 16 kHz mono, 4 s clips.
- 128-bin log-Mel spectrogram.
- 25 ms window, 10 ms hop.
- Zero-mean, unit-variance.
- One pipeline, all models.

Figure 1. *End-to-end detection pipeline.*

Note. *Same preprocessing path for all models.*

Methodology — Architecture and Training

CNN + Squeeze-and-Excitation Model

Stage	Size	Role
Input	1 x 128 x 401	normalized log-Mel spectrogram
Conv block 1	32 channels	two 3x3 convs + BN/ReLU + max pool
Conv block 2	64 channels	two 3x3 convs + BN/ReLU + max pool
Conv block 3	128 channels	two 3x3 convs + BN/ReLU + max pool
Conv block 4	256 channels	two 3x3 convs + BN/ReLU + max pool
SE block	r = 16	global pooling + channel recalibration
Classifier	512 -> 1	dropout + binary logit

Figure 2. *CNN + SE architecture (1.31 M params).*

- 4 conv blocks: 32 → 256 ch.
- SE attention, ratio r = 16.
- GAP → FC 512 → 1 logit.
- Adam, LR 1e-4, cosine.
- BCE with class weights.
- 30 epochs, best-EER pick.

Dataset Overview

Table 1

ASVspoof Logical Access splits used in this study.

Split	Bonafide	Spoof	Total	Role
2019 LA train	2,580	22,800	25,380	Model fitting
2019 LA dev	2,548	22,296	24,844	Checkpoint selection
2021 LA eval	18,452	163,114	181,566	Final evaluation

Note. *2021 LA evaluation includes spoofing systems A07–A19, none of which appear in training.*

- Spoof outnumbers bonafide ~9:1; loss is class-weighted.

Results I — Main Metrics

Table 2

Main metrics on full 2021 LA evaluation.

Metric	Value
Equal Error Rate (EER)	18.28%
AUC-ROC	0.8979
Accuracy @ 0.5	0.9072
Precision @ 0.5	0.9567
Recall @ 0.5	0.9392
F1-score @ 0.5	0.9479
Latency per 4 s clip (MPS)	1.66 ms

Note. *n = 181,566. EER and AUC use continuous scores.*

- EER above 5% target.
- Strong F1 from class imbalance.
- Real-time latency on laptop.
- Useful pipeline, not deployable.

Results II — ROC and Confusion Matrix

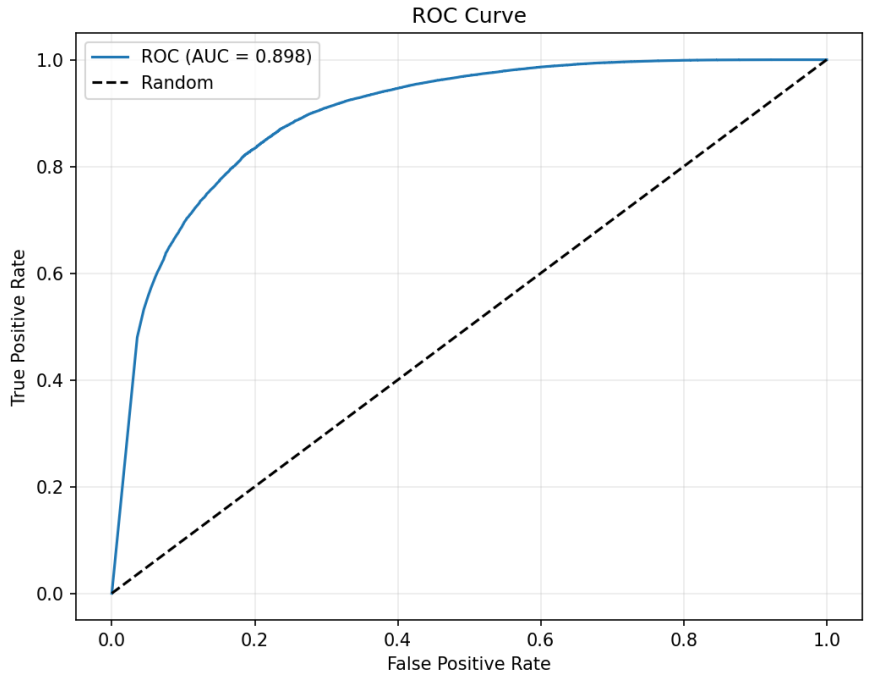


Figure 3. *ROC curve, 2021 LA evaluation.*

Note. *AUC-ROC = 0.8979.*

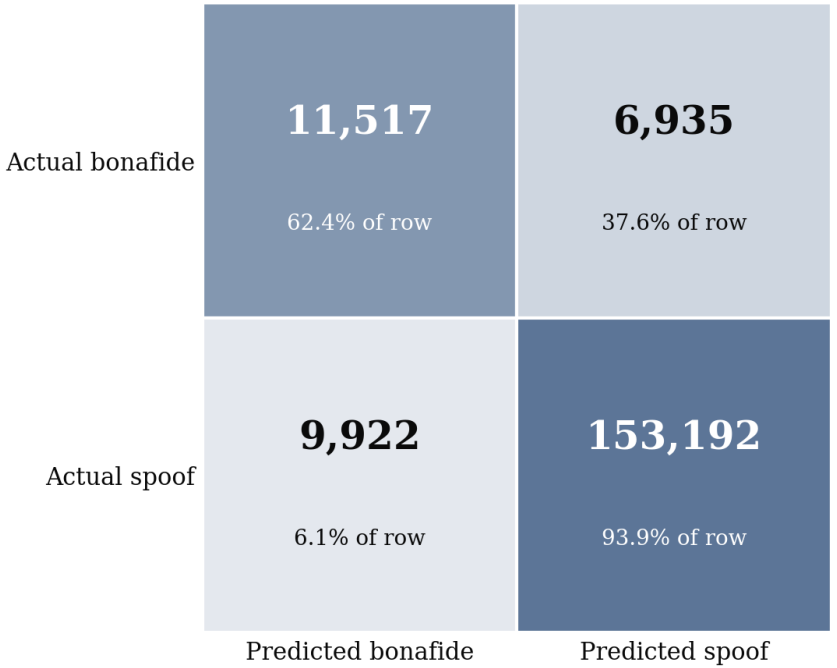


Figure 4. *Confusion matrix at threshold 0.5.*

Note. *False-positive rate 37.6% indicates miscalibration.*

Results III — Model Comparison

Table 3

Model comparison on full 2021 LA evaluation (n = 181,566).

Model	EER	AUC	F1	Params
CNN + SE (main)	18.28%	0.898	0.948	1.31 M
ResNet-18 (baseline)	17.21%	0.896	0.915	11.17 M
LFCC-GMM (baseline)	30.89%	0.740	0.813	n / a

Note. *Latency: CNN + SE 1.66 ms, ResNet-18 2.23 ms per 4 s clip on Apple Silicon MPS.*

- ResNet-18 best EER; CNN + SE 8× smaller.
- LFCC-GMM far weaker, ~13 EER points worse.

Robustness — Stress Test and Cross-Vocoder

Table 4

Robustness evaluation under stressed and held-out conditions.

Condition	EER	AUC	Accuracy
Clean 2021 LA	18.28%	0.8979	0.9072
Podcast-augmented 2021 LA	18.48%	0.8852	0.8991
Cross-vocoder hold-out A05/A06	5.07%	0.9881	0.4131

Note. *Cross-vocoder accuracy is low at threshold 0.5 due to score-distribution shift; AUC remains high.*

- Augmentation: EER stable, AUC drops slightly.
- Cross-vocoder ranking strong; calibration breaks.

Discussion and Limitations

- Did not meet 5% EER starting goal.
- Threshold 0.5 mis-calibrated for distribution.
- Augmentation synthetic, not real DAW chain.
- Latency is a single-laptop measurement.
- ResNet-18 lower EER, much larger model.
- Pipeline is sound; numbers are honest.

Conclusion and Future Work

Key takeaways

- Full ASVspoof pipeline runs end-to-end.
- Beat LFCC-GMM by ~13 EER points.
- Competitive with ResNet-18, far smaller.

Future work

- Threshold calibration on production data.
- Real DAW export-chain testing.
- Stronger architectures: RawNet, transformer.